# Centrifuge



Category: **Laboratory Instruments & Glassware** Type: **Full SOP** Priority: **High** 4 labs

**Covers these lab items:** Centrifuge

## What it is used for

A centrifuge separates components of a sample by spinning it at a controlled speed. Correct balancing, compatible tubes, and proper rotor handling help produce reliable results and reduce the risk of spills or mechanical failure.

## Main parts

- Rotor holds the sample tubes or containers during rotation.
- Lid and locking mechanism secure the chamber during operation.
- Drive motor rotates the rotor at the selected speed.
- Speed and time controls set the operating cycle.
- Display shows speed, time, and operating status on equipped models.
- Tube holders or adapters support compatible sample containers.
- Safety interlocks prevent operation when the lid is not properly secured.
- Chamber contains the rotor and helps limit exposure to moving parts.

## Before you start

1. Only trained personnel should operate the centrifuge under the relevant laboratory procedure.
2. Read the manual and confirm the rotor, tubes, and adapters are compatible with the sample and speed.
3. Inspect the rotor, chamber, lid, seals, and tubes for cracks, corrosion, or damage.
4. Balance sample tubes by matching their mass and positions according to the rotor instructions.
5. Confirm the sample containers are closed and suitable for centrifugation.
6. Set the centrifuge on a stable, level surface and check that ventilation openings are clear.

## How to use

1. Switch off the centrifuge before installing or removing the rotor or adapters.
2. Load balanced tubes into the correct rotor positions and secure them as instructed.
3. Close the lid fully and confirm the locking mechanism engages.
4. Set the required speed, time, and temperature if the model supports temperature control.
5. Verify the settings against the approved method before starting the run.
6. Start the centrifuge and observe it from a safe distance for unusual vibration or noise.
7. Stop the run immediately if severe vibration, unusual noise, or a warning occurs, following the manual.
8. Wait until the rotor has completely stopped before opening the lid.

9. Remove tubes carefully and inspect them for leakage, breakage, or abnormal sample appearance.
10. Clean the chamber and accessories as required and record the run or any observed fault.

## ⚠ Safety

- Never open the lid or attempt to slow the rotor by hand while it is spinning.
- Do not operate the centrifuge with unbalanced loads, damaged rotors, or cracked tubes.
- Use only the rotor speed and tube ratings approved by the manufacturer.
- Handle potentially infectious or hazardous samples according to the laboratory's containment procedures.
- If a tube breaks or leaks, keep the lid closed and follow the laboratory's spill response procedure.
- Disconnect power before maintenance and never bypass the lid interlock.

## Care & storage

- Clean the chamber, rotor, and adapters using materials compatible with the equipment and sample type.
- Inspect the rotor regularly for corrosion, cracks, and wear, and remove damaged parts from service.
- Dry components as instructed and store the rotor in the recommended position.
- Maintain service, inspection, and rotor-use records according to laboratory procedures.

## Common mistakes to avoid

- Loading tubes without balancing their mass and positions.
- Using tubes or adapters that are incompatible with the rotor or sample.
- Opening the lid before the rotor has completely stopped.
- Ignoring unusual vibration, noise, leakage, or rotor damage.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*